

Units, Physical quantities, and vectors



- Predicting the path of a thunderstorm helps reduce damage to life and property.
- The thunderstorm is moving with a speed of 20 km/h.
- Its direction is 53° north of east.
- The time considered is 1 hour.
- The problem asks for the distance moved toward the north in 1 hour.

Physics

- Physics is one of the most fundamental sciences.
- Ideas of physics are used by scientists in many fields such as chemistry, paleontology, and climatology.
- Physics forms the foundation of all engineering and technology.
- Engineering designs (TVs, spacecraft, tools, etc.) require understanding basic physical laws.
- Studying physics is challenging but also rewarding and intellectually satisfying.
- Physics explains everyday and natural phenomena (blue sky, radio waves, satellite motion).
- Physics represents a major achievement of human intellect in understanding nature.
- The chapter introduces essential preliminary concepts for studying physics.
- Topics include:
 - Nature of physical theories
 - Use of idealized models
 - Systems of units
 - Accuracy and uncertainty in measurements
 - Use of rough estimates when exact answers are not practical
 - Basics of vectors and vector algebra. Vectors are important for describing physical quantities with both magnitude and direction (e.g., velocity, force).

The Nature of Physics

- Physics is an experimental science.
- Physicists observe natural phenomena.
- They look for patterns that relate different phenomena.
- These patterns are known as physical theories.
- Well-established and widely accepted theories are called physical laws or principles.

Caution: The meaning of the word “theory

- A theory is not a random or unproven idea.
- It is a well-supported explanation of natural phenomena.
- Theories are based on observations and accepted fundamental principles.
- A scientific theory is developed through extensive research.
- Example: Theory of biological evolution, established through generations of observation and study.

The Nature of Physics

- Developing a physical theory involves:
- Asking appropriate questions.
- Designing experiments.
- Drawing correct conclusions from results.
- Galileo Galilei studied the motion of falling objects.
- He concluded that the acceleration of a falling body is independent of its weight.
- The development of physical theories is often indirect and includes mistakes and revisions.
- Physics is not just a set of facts, but a process of discovering general principles.

- No physical theory is considered absolutely final or perfect.
- Conflicting observations can disprove a theory, but it can never be proven completely true.
- Galileo's theory appeared incorrect when comparing a feather and a cannonball in air.
- The difference in falling rates is due to air resistance and buoyancy, not weight.
- In a vacuum, both objects fall at the same rate, confirming Galileo's idea.
- Every theory has a limited range of validity.
- Galileo's theory applies when air forces are much smaller than the object's weight.
- Later theories can extend the range of validity of earlier ones.
- Newton's laws expanded and improved Galileo's work on falling bodies.

Standards and units

Fundamental Quantities	SI Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of substance	mol	mol
Luminous Intensity	candela	cd

Unit Prefixes

Submult	Prefix	Char	Mult	Prefix	Char
10^{-1}	deci	d	10	deca	da
10^{-2}	centi	c	10^2	hecto	h
10^{-3}	milli	m	10^3	kilo	k
10^{-6}	micro	u	10^6	mega	M
10^{-9}	nano	n	10^9	giga	G
10^{-12}	pico	p	10^{12}	tera	T
10^{-15}	femto	f	10^{15}	peta	P
10^{-18}	atto	a	10^{18}	exa	E
10^{-21}	zepto	z	10^{21}	zetta	Z
10^{-24}	yocto	y	10^{24}	yotta	Y

1 kilometer = 1 km = 10^3 meters = 10^3 m

1 kilogram = 1 kg = 10^3 grams = 10^3 g

1 kilowatt = 1 kW = 10^3 watts = 10^3 W

Some typical lengths in the universe



(a) 10^{26} m
Limit of the
observable
universe



(b) 10^{11} m
Distance to
the sun



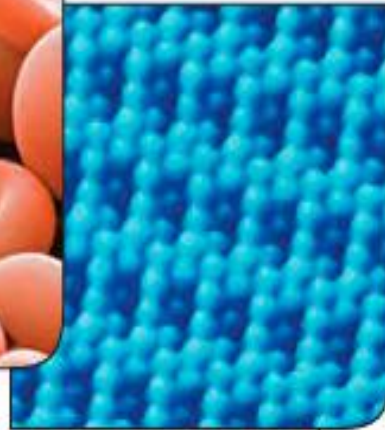
(c) 10^7 m
Diameter of
the earth



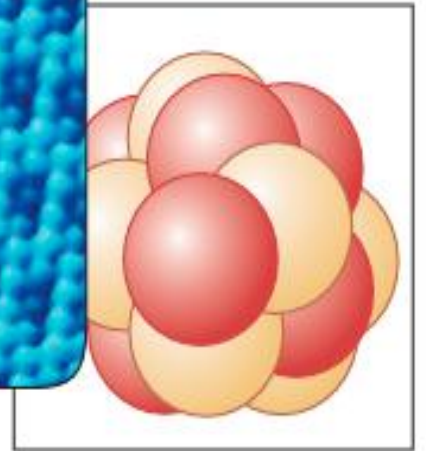
(d) 1 m
Human
dimensions



(e) 10^{-5} m
Diameter of a
red blood cell



(f) 10^{-10} m
Radius of an
atom



(g) 10^{-14} m
Radius of an
atomic nucleus

position

The position of an object tells us where the object is located in space at a particular time. To define a position clearly and unambiguously, we always need a reference.

The position of an object is its location relative to a chosen reference point, called the origin.

Definition:

Position is the location of an object with respect to a fixed reference point in a chosen coordinate system.

◆ Example:

Saying “The book is on the table” is not enough.

We must say, “The book is 50 cm to the right of the table’s center.”

The table’s center acts as the reference point (origin).

Coordinate System

To define position precisely, we use a coordinate system.

➤ (a) One-Dimensional (1D) Position

Used when motion is along a straight line (e.g., train on a track).

A straight line is chosen

One point is taken as the origin (O)

Position is given by a single number (x)

Example:

$x = +5 \text{ m} \rightarrow$ object is 5 m to the right of the origin.

$x = -3 \text{ m} \rightarrow$ object is 3 m to the left of the origin.

➤ Two-Dimensional (2D) Position

Used for motion in a plane (e.g., car on a road map).

Two coordinates give a position:

x-coordinate

y-coordinate

Example:

Position = (3 m, 4 m)

➤ Three-Dimensional (3D) Position

Used in space (e.g., flying bird, satellite).

Position is given by:

x, y, z coordinates

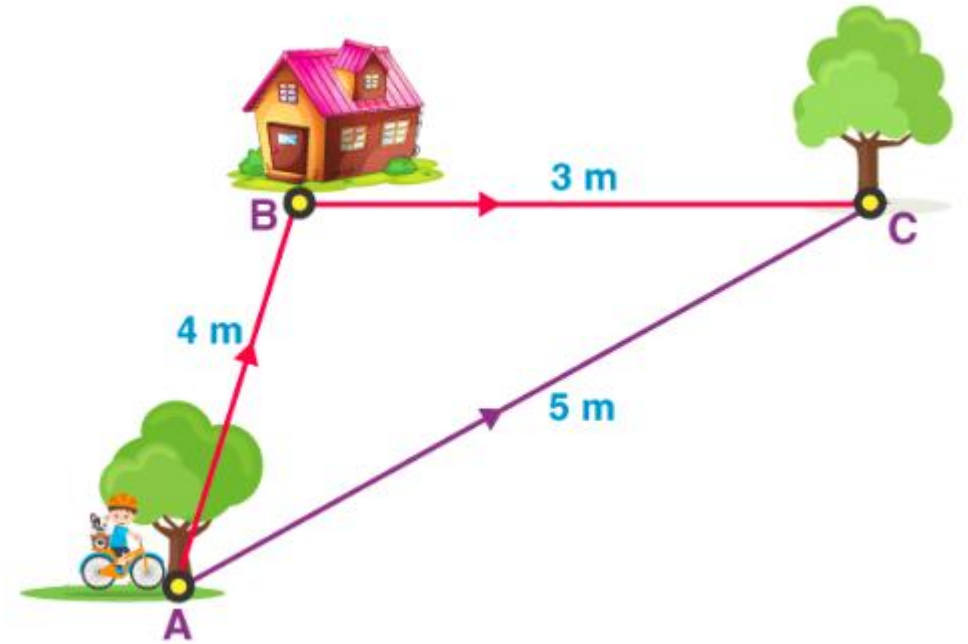
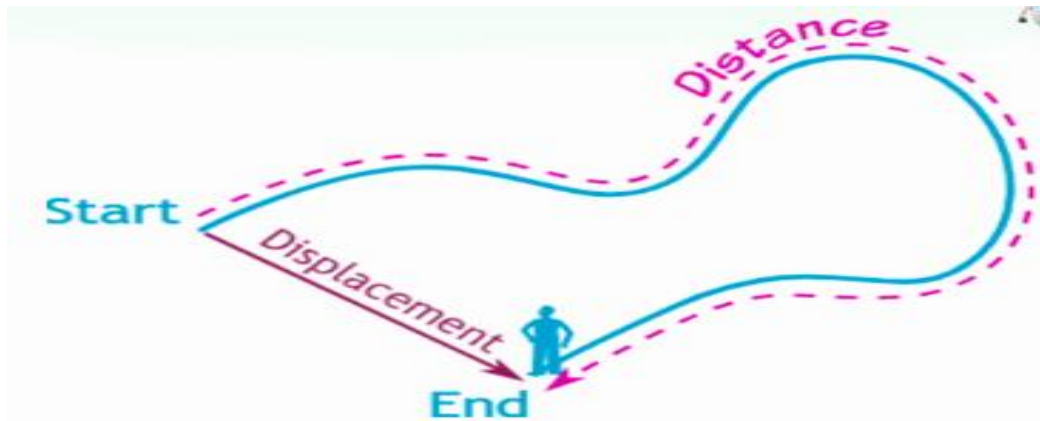
Example:

Position = (2 m, -1 m, 5 m)

Distance and Displacement

- Displacement refers to the change in position of an object or the distance and direction from its initial point to its final point.
- Distance involves the total ground covered, regardless of direction.

Distance and displacement are two quantities that seem to mean the same, but are distinctly different with different meanings and definitions. Distance is the measure of “how much ground an object has covered during its motion,” while displacement refers to the measure of “how far out of place is an object.”



Distance from A to C = $4\text{m} + 3\text{m} = 7\text{m}$
Displacement = 5m

Displacement is the change in position of an object in a particular direction.

Mathematically:

Displacement = Final Position – Initial Position

➤ Vector Nature of Displacement

Displacement is a vector quantity, which means:

It has a magnitude.

It has direction.

This makes displacement different from distance.

➤ Displacement Using Position Vectors

$$\vec{d} = \vec{r}_f - \vec{r}_i$$

Initial position vector = $\vec{r}_i$

Final position vector = $\vec{r}_f$

Average velocity

- Average velocity describes how fast and in which direction an object's position changes over a given time interval.
- Average velocity is defined as the total displacement divided by the total time taken.

Mathematically:

$$\text{Average Velocity} = \frac{\text{Total displacement}}{\text{Total time}} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

Initial position vector = $\vec{r}_i$

Final position vector = $\vec{r}_f$

Total time taken = Δt

$$\vec{v}_{avg} = \frac{\vec{r}_f - \vec{r}_i}{\Delta t}$$

Direction of Average Velocity

- The direction of average velocity is the same as the direction of displacement.
- If displacement is zero, average velocity is zero, even if the object was moving.

- **CAUTION** The meaning of Δx

- Note that Δx is not the product of Δ and x ; it is a single symbol that means “the change in the quantity x .” We always use the Greek capital letter (delta) to represent a change in a quantity, equal to the final value of the quantity minus the initial value — never the reverse. Likewise, the time interval from t_1 to t_2 change in the quantity t , $\Delta t = t_2 - t_1$ (final time minus initial time).

The Acceleration vector

Acceleration is the **rate of change of velocity with time**.

Since **velocity is a vector**, **acceleration is also a vector**.

$$\vec{a}_{\text{av}} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t} \quad (\text{average acceleration vector})$$

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \quad (\text{instantaneous acceleration vector})$$

$$a_x = \frac{dv_x}{dt} \quad a_y = \frac{dv_y}{dt} \quad a_z = \frac{dv_z}{dt} \quad (\text{components of instantaneous acceleration})$$

$$a_x = \frac{d^2x}{dt^2} \quad a_y = \frac{d^2y}{dt^2} \quad a_z = \frac{d^2z}{dt^2}$$

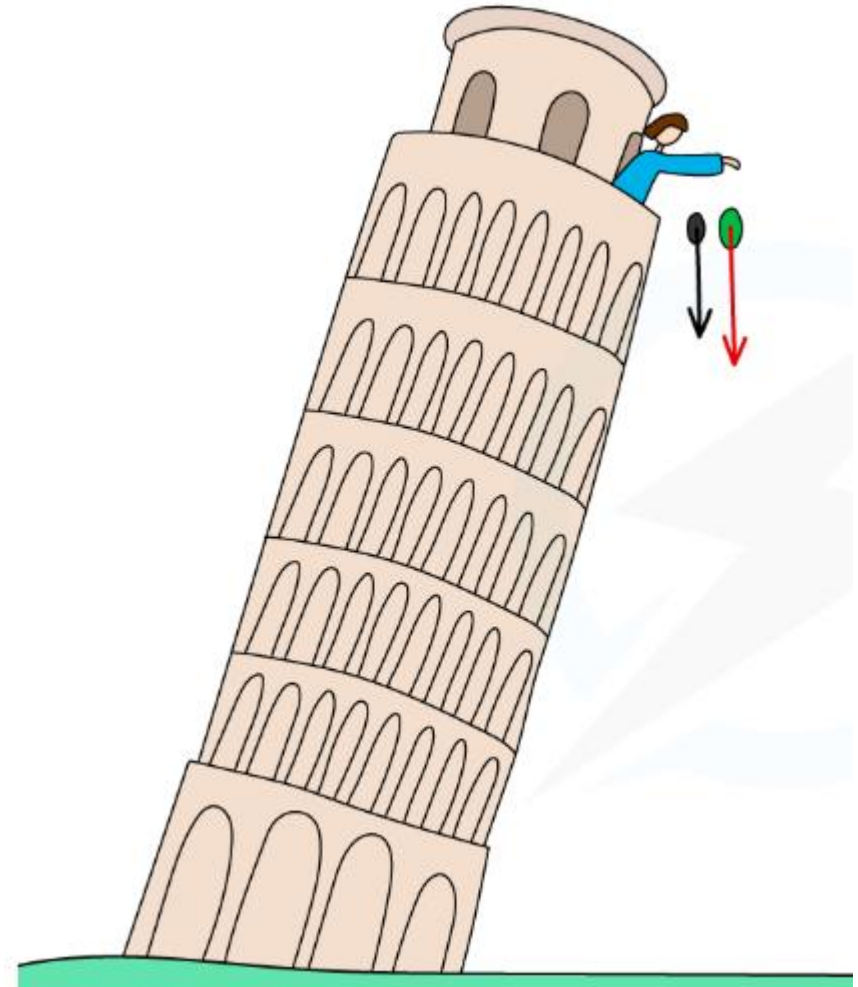
$$\vec{a} = \frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j} + \frac{d^2z}{dt^2} \hat{k}$$

In terms of unit vectors

$$\vec{a} = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k}$$

Free-fall acceleration

- Free fall acceleration, or g , is the constant downward acceleration of objects due to gravity, approximately 9.8 m/s^2 (32 ft/s^2) at Earth's surface, meaning velocity increases by this amount each second.
- In a vacuum, all objects fall at this rate regardless of mass, but air resistance affects real-world objects, slowing them down and eventually leading to terminal velocity.



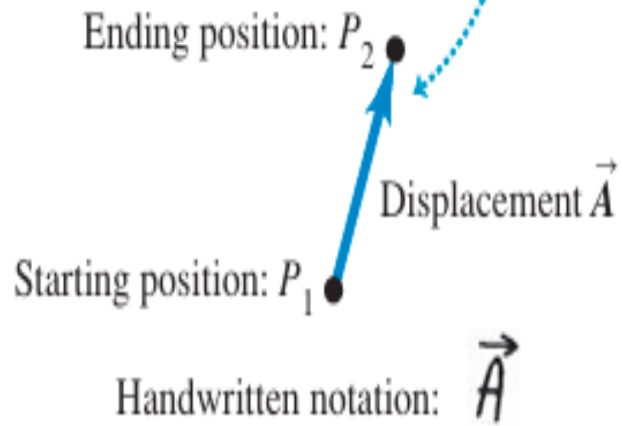
Normal Equations	Same equations specialized for freefall $\mathbf{a} \rightarrow \mathbf{g}$ and $\mathbf{x} \rightarrow \mathbf{y}$
$v_f = v_i + at$ $v_f^2 = v_i^2 + 2a\Delta x$ $\Delta x = v_i t + 1/2 at^2$	$v_{fy} = v_{iy} + gt$ $v_{fy}^2 = v_{iy}^2 + 2g\Delta y$ $\Delta y = v_{iy} t + 1/2 gt^2$

Vectors

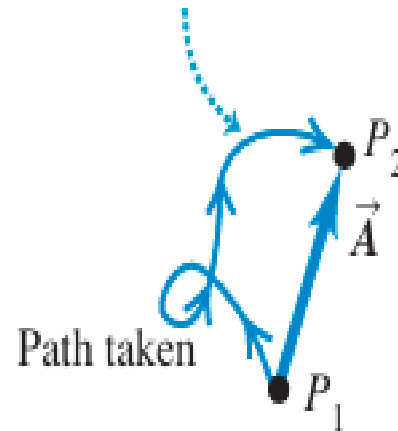
Scalar Quantity	Vector Quantity
1. Scalar quantity has only magnitude, but no direction. 2. Every scalar quantity is one dimensional. 3. Any change in scalar quantity is the reflection of change in magnitude. 4. Scalar quantity cannot be resolved as it has exactly same value regardless of direction. 5. Any mathematical operation carried out among two or more scalar quantities will provide a scalar only. However, if a scalar is operated with a vector then the result will be a vector. 6. Few examples of scalar quantity: length, mass, energy, density, etc.	1. Vector quantity has both magnitude and direction. 2. Vector quantity can be 1-D, 2-D or 3-D. 3. Any change in vector quantity can reflect either change in direction or change in magnitude or changes in both. 4. Vector can be resolved in any direction using sine or cosine of the adjacent angle. 5. Result of mathematical operations between two or more vectors may give either scalar or vector. Dot product of two vectors gives only scalar; while, cross product or summation or subtraction results in a vector. 6. Examples of vector quantity: displacement, velocity, acceleration, force, etc.

- Displacement as a vector quantity. A displacement is always a straight-line segment directed from the starting point to the ending point, even if the path is curved.

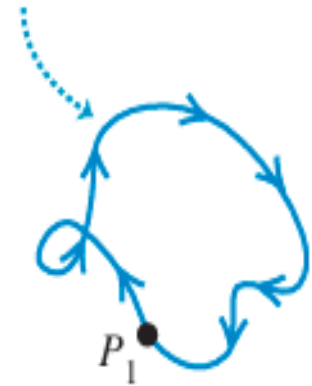
(a) We represent a displacement by an arrow pointing in the direction of displacement.



(b) Displacement depends only on the starting and ending positions—not on the path taken.



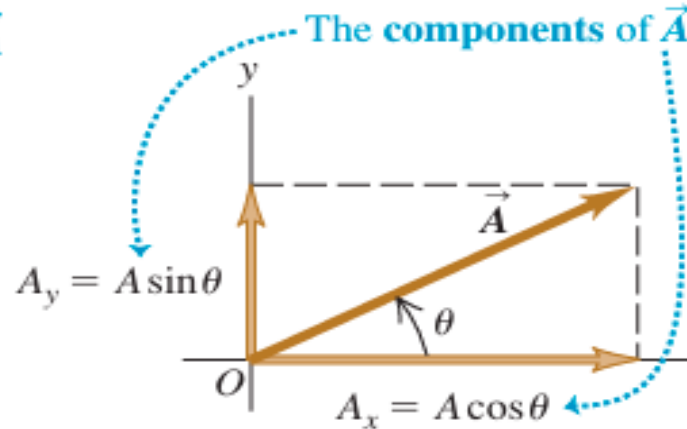
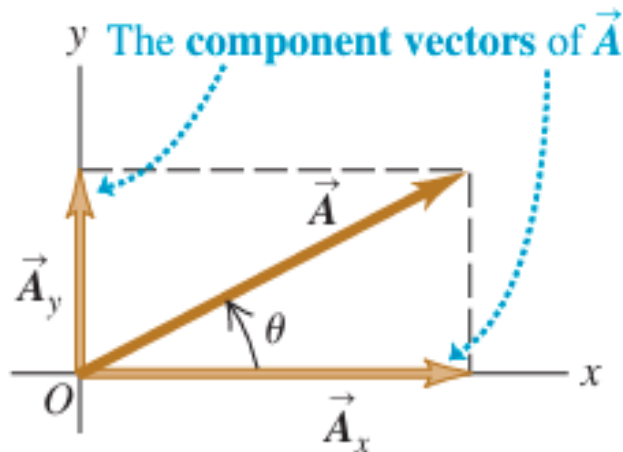
(c) Total displacement for a round trip is 0, regardless of the distance traveled.



Components of Vectors

- Components of a vector are the parts of a vector along chosen coordinate axes (usually the x-axis and y-axis).
- A single vector can be replaced by two or three perpendicular vectors whose combined effect is the same as the original vector.

CAUTION: *Components are not vectors* The components A_x and A_y of a vector $\vec{A}$ are just numbers; they are not vectors themselves. This is why we print the symbols for components in light italic type with no arrow on top instead of in boldface italic with an arrow, which is reserved for vectors.



$$\vec{A} = \vec{A}_x + \vec{A}_y$$

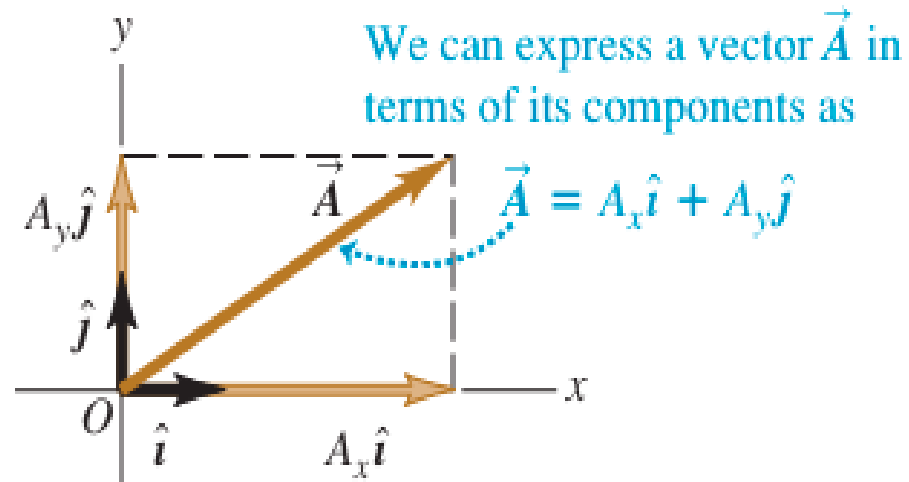
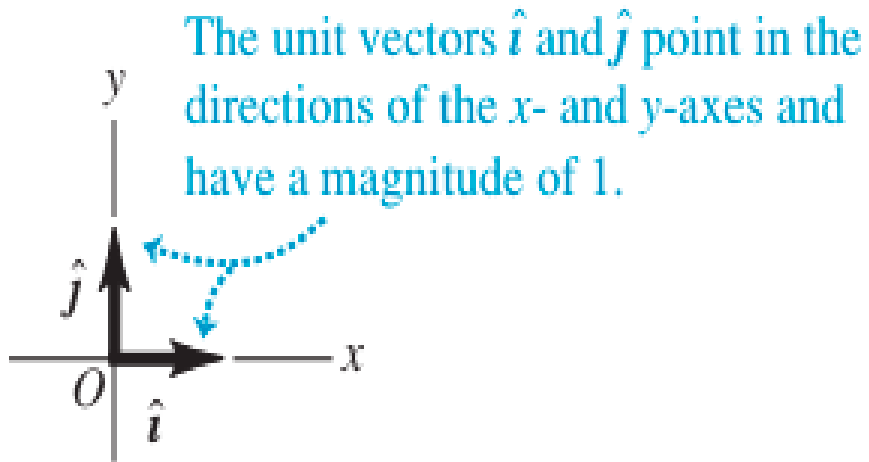
$$\frac{A_x}{A} = \cos \theta \quad \text{and} \quad \frac{A_y}{A} = \sin \theta$$

$$A_x = A \cos \theta \quad \text{and} \quad A_y = A \sin \theta$$

(θ measured from the $+x$ -axis, rotating toward the $+y$ -axis)

unit vector

- A unit vector is a vector that has a magnitude of 1, with no units.
- Its only purpose is to point—that is, to describe a direction in space.
- Unit vectors provide a convenient notation for many expressions involving components of vectors.
- We will always include a caret or “hat” in the symbol for a unit vector to distinguish it from ordinary vectors whose magnitude may or may not be equal to 1.



$$\vec{A}_x = A_x \hat{i}$$
$$\vec{A}_y = A_y \hat{j}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j}$$

$$\vec{R} = \vec{A} + \vec{B}$$

$$= (A_x \hat{i} + A_y \hat{j}) + (B_x \hat{i} + B_y \hat{j})$$

$$= (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j}$$

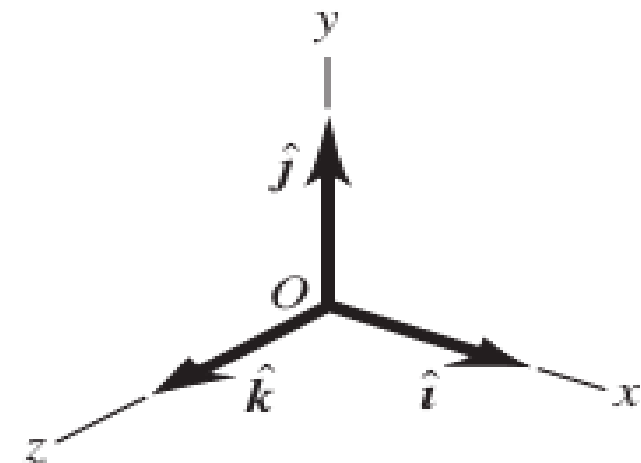
$$= R_x \hat{i} + R_y \hat{j}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{R} = (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j} + (A_z + B_z) \hat{k}$$

$$= R_x \hat{i} + R_y \hat{j} + R_z \hat{k}$$



Example

Given the two displacements

$$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) \text{ m} \quad \text{and}$$

$$\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) \text{ m}$$

find the magnitude of the displacement $2\vec{D} - \vec{E}$.

$$\begin{aligned}\vec{F} &= 2(6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) \text{ m} - (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) \text{ m} \\ &= [(12.00 - 4.00)\hat{i} + (6.00 + 5.00)\hat{j} + (-2.00 - 8.00)\hat{k}] \text{ m} \\ &= (8.00\hat{i} + 11.00\hat{j} - 10.00\hat{k}) \text{ m}\end{aligned}$$

From Eq. (1.12) the magnitude of $\vec{F}$ is

$$\begin{aligned}F &= \sqrt{F_x^2 + F_y^2 + F_z^2} \\ &= \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} \\ &= 16.9 \text{ m}\end{aligned}$$

Arrange the following vectors in order of their magnitude, with the vector of largest magnitude first.

(i) $A = (3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}) \text{ m}$

(ii) $B = (-3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}) \text{ m}$

(iii) $C = (3\mathbf{i} - 5\mathbf{j} - 2\mathbf{k}) \text{ m}$

(iv) $D = (3\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}) \text{ m}$.

To arrange the vectors by magnitude, we calculate the magnitude of each vector using:

$$|\vec{V}| = \sqrt{x^2 + y^2 + z^2}$$

$$\vec{C} = (3\mathbf{i} - 5\mathbf{j} - 2\mathbf{k})$$

$$|\vec{C}| = \sqrt{3^2 + (-5)^2 + (-2)^2} = \sqrt{38}$$

$$\vec{A} = (3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k})$$

$$|\vec{A}| = \sqrt{3^2 + 5^2 + (-2)^2} = \sqrt{9 + 25 + 4} = \sqrt{38}$$

$$\vec{D} = (3\mathbf{i} + 5\mathbf{j} + 2\mathbf{k})$$

$$|\vec{D}| = \sqrt{3^2 + 5^2 + 2^2} = \sqrt{38}$$

$$\vec{B} = (-3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k})$$

$$|\vec{B}| = \sqrt{(-3)^2 + 5^2 + (-2)^2} = \sqrt{38}$$

$$A = B = C = D$$

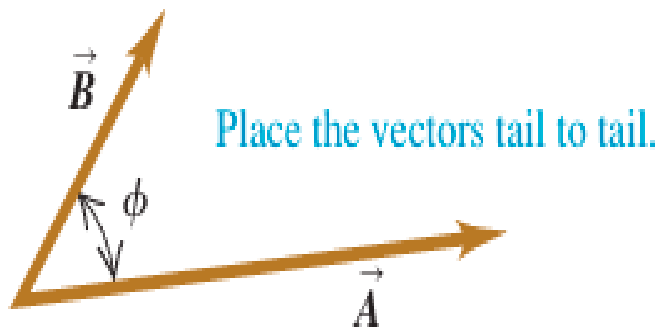
All four vectors have **equal magnitude**.

Product of vectors

Scalar Product

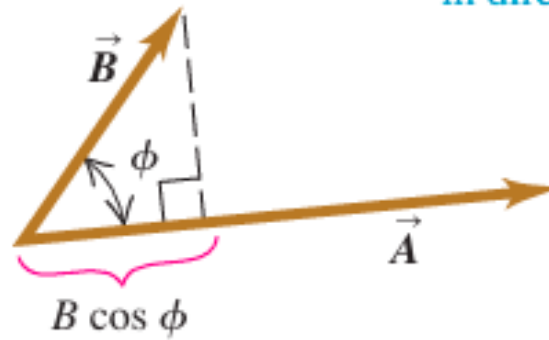
The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is denoted By $\vec{A} \cdot \vec{B}$. Because of this notation, the scalar product is also called the dot product. Although $\vec{A}$ and $\vec{B}$ are vectors, the quantity $\vec{A} \cdot \vec{B}$ is a scalar.

(a)



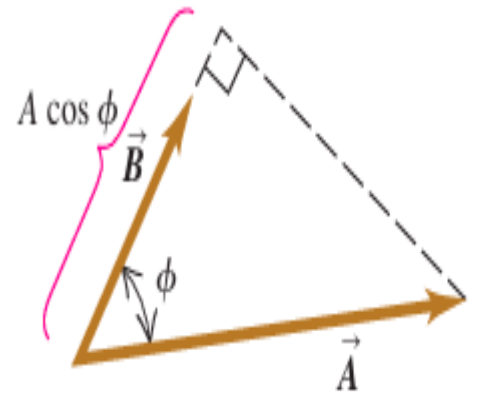
(b) $\vec{A} \cdot \vec{B}$ equals $A(B \cos \phi)$.

(Magnitude of $\vec{A}$) times (Component of $\vec{B}$ in direction of $\vec{A}$)



(c) $\vec{A} \cdot \vec{B}$ also equals $B(A \cos \phi)$

(Magnitude of $\vec{B}$) times (Component of $\vec{A}$ in direction of $\vec{B}$)



➤ To define the scalar product $\vec{A} \cdot \vec{B}$ we draw the two vectors $\vec{A}$ and $\vec{B}$ with their tails at the same point. The angle ϕ (the Greek letter phi) between their directions ranges from 0° to 180° . The projection of the vector onto the direction of this projection is the component of in the direction of and is equal to (We can take components along any direction that's convenient, not just the x- and y-axes.) We define to be the magnitude of multiplied by the component of $\vec{B}$ in the direction of $\vec{A}$.

$$\vec{A} \cdot \vec{B} = AB \cos \phi = |\vec{A}| |\vec{B}| \cos \phi \quad \begin{array}{l} \text{(definition of the scalar} \\ \text{(dot) product)} \end{array}$$

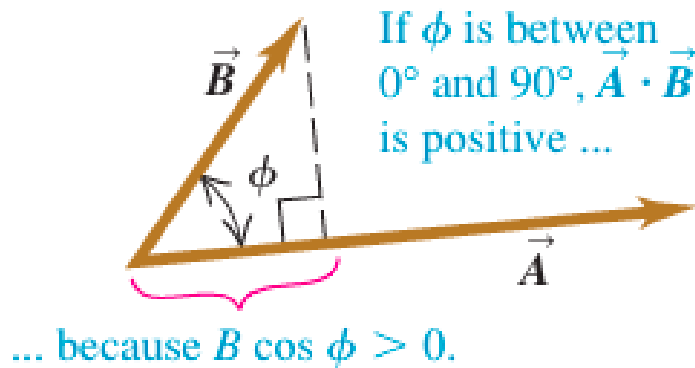
The scalar product is a scalar quantity, not a vector, and it may be positive, negative, or zero. When ϕ is between 0° and 90° , $\cos \phi > 0$ and the scalar product is positive.

When ϕ is between 90° and 180° so that $\cos \phi < 0$, the component of $\vec{B}$ in the direction of $\vec{A}$ is negative, and $\vec{A} \cdot \vec{B}$ is negative.

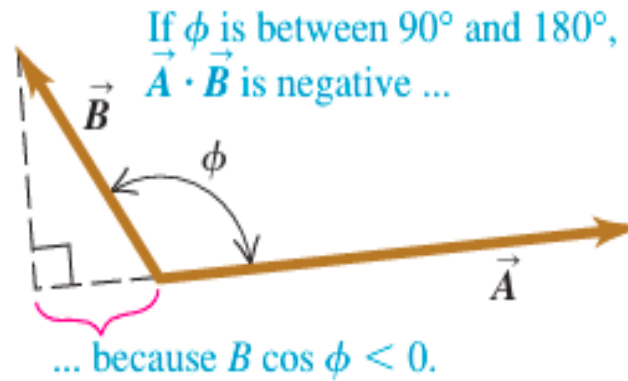
Finally, when $\phi = 90^\circ$, $\vec{A} \cdot \vec{B} = 0$. The scalar product of two perpendicular vectors is always zero.

For any two vectors $\vec{A}$ and $\vec{B}$, $AB \cos \phi = BA \cos \phi$. This means that $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$. The scalar product obeys the commutative law of multiplication; the order of the two vectors does not matter.

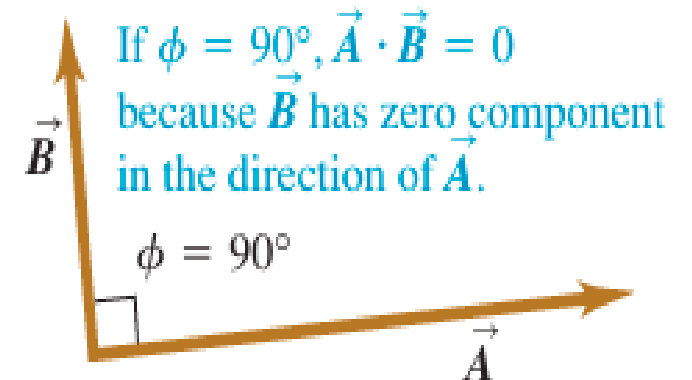
(a)



(b)



(c)



Calculating the Scalar Product Using Components

$$\begin{aligned}\vec{A} \cdot \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\&= A_x \hat{i} \cdot B_x \hat{i} + A_x \hat{i} \cdot B_y \hat{j} + A_x \hat{i} \cdot B_z \hat{k} \\&\quad + A_y \hat{j} \cdot B_x \hat{i} + A_y \hat{j} \cdot B_y \hat{j} + A_y \hat{j} \cdot B_z \hat{k} \\&\quad + A_z \hat{k} \cdot B_x \hat{i} + A_z \hat{k} \cdot B_y \hat{j} + A_z \hat{k} \cdot B_z \hat{k} \\&= A_x B_x \hat{i} \cdot \hat{i} + A_x B_y \hat{i} \cdot \hat{j} + A_x B_z \hat{i} \cdot \hat{k} \\&\quad + A_y B_x \hat{j} \cdot \hat{i} + A_y B_y \hat{j} \cdot \hat{j} + A_y B_z \hat{j} \cdot \hat{k} \\&\quad + A_z B_x \hat{k} \cdot \hat{i} + A_z B_y \hat{k} \cdot \hat{j} + A_z B_z \hat{k} \cdot \hat{k}\end{aligned}$$

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = (1)(1) \cos 0^\circ = 1$$

$$\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = (1)(1) \cos 90^\circ = 0$$

Thus, the scalar product of two vectors is the sum of the products of their respective components.

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z \quad \text{(scalar (dot) product in terms of components)}$$

Finding an angle with the scalar product

Find the angle between the vectors

$$\begin{aligned}\vec{A} &= 2.00\hat{i} + 3.00\hat{j} + 1.00\hat{k} \quad \text{and} \\ \vec{B} &= -4.00\hat{i} + 2.00\hat{j} - 1.00\hat{k}\end{aligned}$$

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{A_x B_x + A_y B_y + A_z B_z}{AB}$$

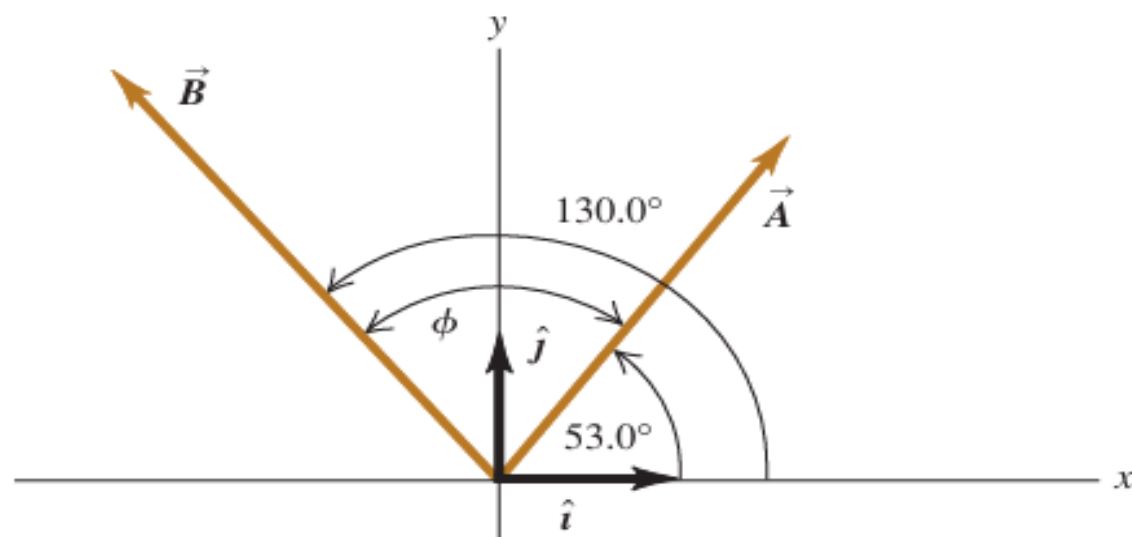
$$\begin{aligned}\vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00) \\ &= -3.00 \\ A &= \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2} \\ &= \sqrt{14.00} \\ B &= \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2} \\ &= \sqrt{21.00} \\ \cos \phi &= \frac{A_x B_x + A_y B_y + A_z B_z}{AB} = \frac{-3.00}{\sqrt{14.00} \sqrt{21.00}} = -0.175 \\ \phi &= 100^\circ\end{aligned}$$

Example 1.10 Calculating a scalar product

Find the scalar product $\vec{A} \cdot \vec{B}$ of the two vectors in Fig. 1.27. The magnitudes of the vectors are $A = 4.00$ and $B = 5.00$.

SOLUTION

IDENTIFY and SET UP: We can calculate the scalar product in two ways: using the magnitudes of the vectors and the angle between them (Eq. 1.18), and using the components of the vectors (Eq. 1.21). We'll do it both ways, and the results will check each other.



EXECUTE: The angle between the two vectors is $\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ$, so Eq. (1.18) gives us

$$\vec{A} \cdot \vec{B} = AB \cos \phi = (4.00)(5.00) \cos 77.0^\circ = 4.50$$

To use Eq. (1.21), we must first find the components of the vectors. The angles of $\vec{A}$ and $\vec{B}$ are given with respect to the $+x$ -axis and are measured in the sense from the $+x$ -axis to the $+y$ -axis, so we can use Eqs. (1.6):

$$A_x = (4.00) \cos 53.0^\circ = 2.407$$

$$A_y = (4.00) \sin 53.0^\circ = 3.195$$

$$B_x = (5.00) \cos 130.0^\circ = -3.214$$

$$B_y = (5.00) \sin 130.0^\circ = 3.830$$

As in Example 1.7, we keep an extra significant figure in the components and round at the end. Equation (1.21) now gives us

$$\begin{aligned}\vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.407)(-3.214) + (3.195)(3.830) + (0)(0) = 4.50\end{aligned}$$

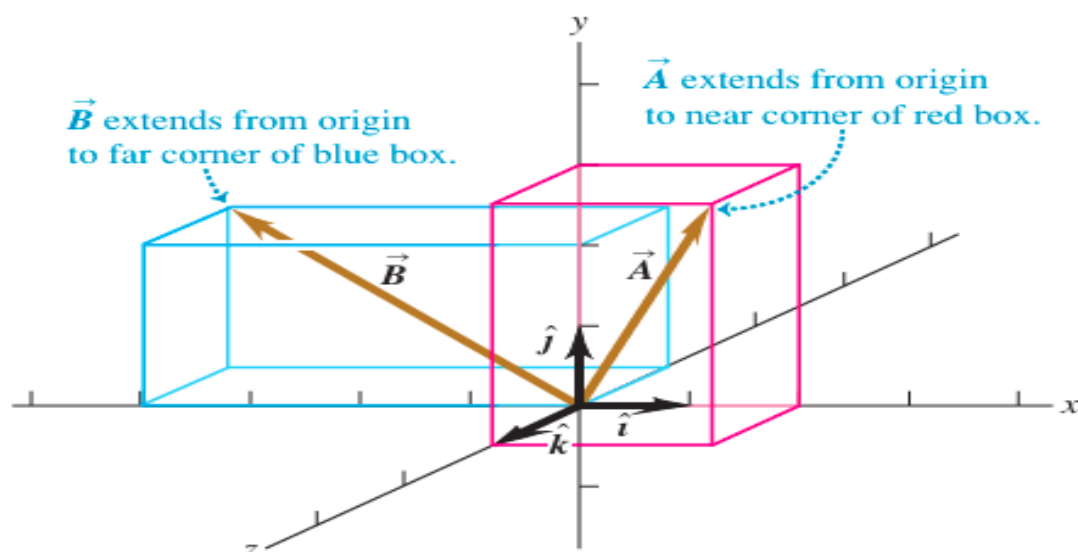
Example 1.11 Finding an angle with the scalar product

Find the angle between the vectors

$$\begin{aligned}\vec{A} &= 2.00\hat{i} + 3.00\hat{j} + 1.00\hat{k} \quad \text{and} \\ \vec{B} &= -4.00\hat{i} + 2.00\hat{j} - 1.00\hat{k}\end{aligned}$$

SOLUTION

IDENTIFY and SET UP: We're given the x -, y -, and z -components of two vectors. Our target variable is the angle ϕ between them (Fig. 1.28). To find this, we'll solve Eq. (1.18), $\vec{A} \cdot \vec{B} = AB \cos \phi$, for ϕ in terms of the scalar product $\vec{A} \cdot \vec{B}$ and the magnitudes A and B . We can evaluate the scalar product using Eq. (1.21),



$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$, and we can find A and B using Eq. (1.7).

EXECUTE: We solve Eq. (1.18) for $\cos \phi$ and write $\vec{A} \cdot \vec{B}$ using Eq. (1.21). Our result is

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{A_x B_x + A_y B_y + A_z B_z}{AB}$$

We can use this formula to find the angle between *any* two vectors $\vec{A}$ and $\vec{B}$. Here we have $A_x = 2.00$, $A_y = 3.00$, and $A_z = 1.00$, and $B_x = -4.00$, $B_y = 2.00$, and $B_z = -1.00$. Thus

$$\begin{aligned}\vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00) \\ &= -3.00 \\ A &= \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2} \\ &= \sqrt{14.00} \\ B &= \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2} \\ &= \sqrt{21.00} \\ \cos \phi &= \frac{A_x B_x + A_y B_y + A_z B_z}{AB} = \frac{-3.00}{\sqrt{14.00} \sqrt{21.00}} = -0.175 \\ \phi &= 100^\circ\end{aligned}$$

Vector Product

- The **vector product** (or **cross product**) of two vectors $\vec{A}$ and $\vec{B}$ is written as $\vec{A} \times \vec{B}$.
- The result of a cross product is **also a vector**.
- It is used to describe physical quantities like **torque**, **angular momentum**, **magnetic fields**, and **magnetic forces**.
- To define the cross product:
 - Place $\vec{A}$ and $\vec{B}$ tail-to-tail; they lie in a plane.
 - The direction of $\vec{A} \times \vec{B}$ is **perpendicular to the plane** containing $\vec{A}$ and $\vec{B}$.
- The **magnitude** of the cross product is:

$$|\vec{A} \times \vec{B}| = AB \sin \phi$$

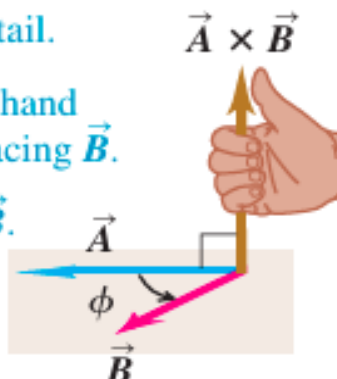
where ϕ is the angle between $\vec{A}$ and $\vec{B}$.

- The angle ϕ is measured from $\vec{A}$ toward $\vec{B}$ and lies between 0° and 180° .
- Since $\sin \phi \geq 0$, the magnitude of the cross product is **never negative**.
- If $\vec{A}$ and $\vec{B}$ are **parallel or antiparallel** ($\phi = 0^\circ$ or 180°):
 - $\sin \phi = 0$
 - The cross product is **zero**.
- The cross product of **any vector with itself is zero**.
- For any plane, there are **two possible perpendicular directions** (one on each side).
- The direction of $\vec{A} \times \vec{B}$ is chosen using the **right-hand rule**.
- **Right-hand rule:**
 - Rotate vector $\vec{A}$ toward $\vec{B}$ through the **smaller angle**.
 - Curl the fingers of your **right hand** in the direction of rotation.
 - The **thumb points in the direction of $\vec{A} \times \vec{B}$** .
- To find the direction of $\vec{B} \times \vec{A}$:
 - Rotate $\vec{B}$ toward $\vec{A}$.
 - The resulting direction is **opposite** to that of $\vec{A} \times \vec{B}$.
- Therefore, the cross product is **not commutative**:

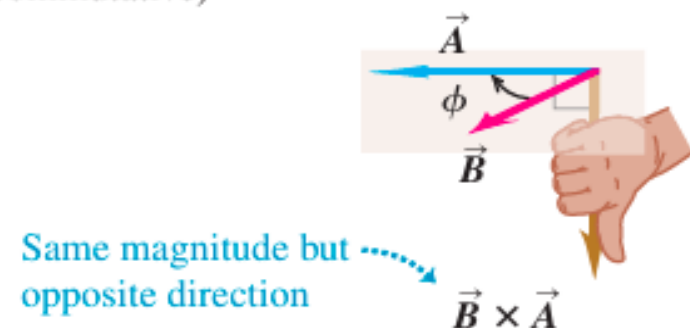
1.29 (a) The vector product $\vec{A} \times \vec{B}$ determined by the right-hand rule.

(b) $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$; the vector product is anticommutative.

(a) Using the right-hand rule to find the direction of $\vec{A} \times \vec{B}$

- ① Place $\vec{A}$ and $\vec{B}$ tail to tail.
 - ② Point fingers of right hand along $\vec{A}$, with palm facing $\vec{B}$.
 - ③ Curl fingers toward $\vec{B}$.
 - ④ Thumb points in direction of $\vec{A} \times \vec{B}$.
- 

(b) $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$ (the vector product is anticommutative)



$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

- Geometrical meaning of magnitude:

- $B \sin \phi$ is the component of $\vec{B}$ perpendicular to $\vec{A}$.
- The magnitude of the cross product is:

$$|\vec{A} \times \vec{B}| = A \times (B \sin \phi)$$

- Similarly:

- The magnitude also equals B times the component of $\vec{A}$ perpendicular to $\vec{B}$.
- This geometric interpretation is valid for **all angles between 0° and 180°** .

Unit vectors product

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = -\hat{j} \times \hat{i} = \hat{k}$$

$$\hat{j} \times \hat{k} = -\hat{k} \times \hat{j} = \hat{i}$$

$$\hat{k} \times \hat{i} = -\hat{i} \times \hat{k} = \hat{j}$$

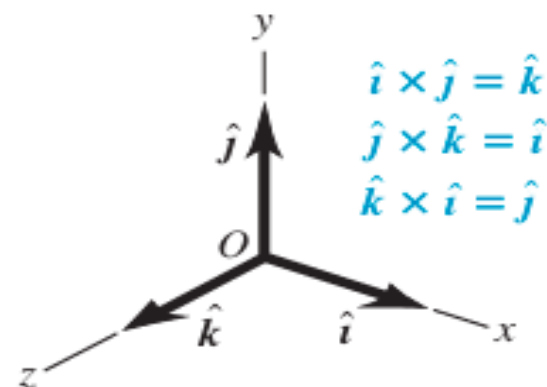
$$\vec{A} \times \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$= A_x \hat{i} \times B_x \hat{i} + A_x \hat{i} \times B_y \hat{j} + A_x \hat{i} \times B_z \hat{k}$$

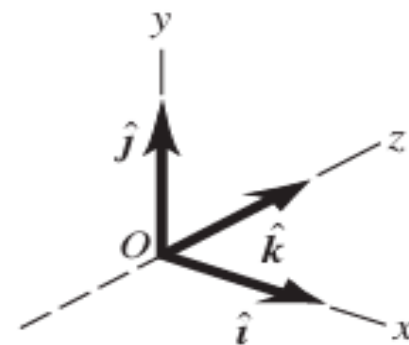
$$+ A_y \hat{j} \times B_x \hat{i} + A_y \hat{j} \times B_y \hat{j} + A_y \hat{j} \times B_z \hat{k}$$

$$+ A_z \hat{k} \times B_x \hat{i} + A_z \hat{k} \times B_y \hat{j} + A_z \hat{k} \times B_z \hat{k}$$

(a) A right-handed coordinate system



(b) A left-handed coordinate system; we will not use these.



$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

Thus the components of $\vec{C} = \vec{A} \times \vec{B}$ are given by

$$C_x = A_y B_z - A_z B_y \quad C_y = A_z B_x - A_x B_z \quad C_z = A_x B_y - A_y B_x$$

(components of $\vec{C} = \vec{A} \times \vec{B}$)

The vector product can also be expressed in determinant form as

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Example 1.12 Calculating a vector product

Vector $\vec{A}$ has magnitude 6 units and is in the direction of the $+x$ -axis. Vector $\vec{B}$ has magnitude 4 units and lies in the xy -plane, making an angle of 30° with the $+x$ -axis (Fig. 1.32). Find the vector product $\vec{C} = \vec{A} \times \vec{B}$.

$$AB \sin \phi = (6)(4)(\sin 30^\circ) = 12$$

By the right-hand rule, the direction of $\vec{A} \times \vec{B}$ is along the $+z$ -axis (the direction of the unit vector $\hat{k}$), so we have $\vec{C} = \vec{A} \times \vec{B} = 12\hat{k}$.

To use Eqs. (1.27), we first determine the components of $\vec{A}$ and $\vec{B}$:

$$A_x = 6$$

$$A_y = 0$$

$$A_z = 0$$

$$B_x = 4 \cos 30^\circ = 2\sqrt{3}$$

$$B_y = 4 \sin 30^\circ = 2$$

$$B_z = 0$$

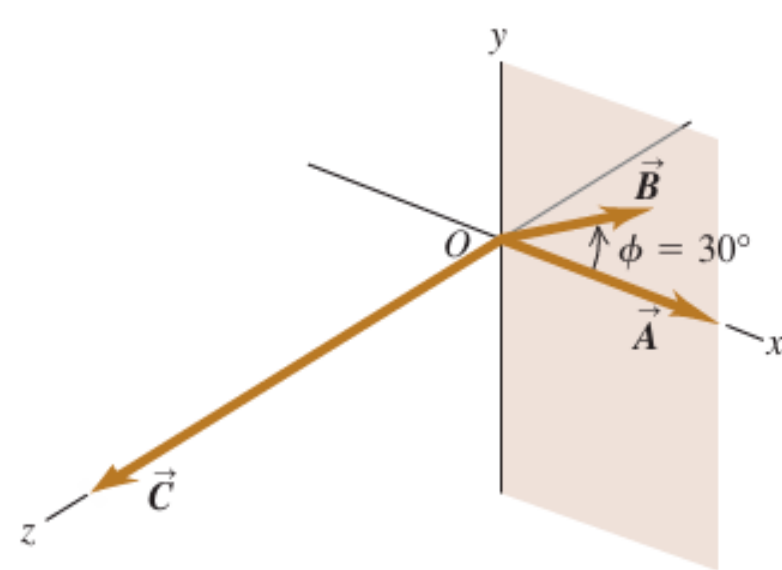
Then Eqs. (1.27) yield

$$C_x = (0)(0) - (0)(2) = 0$$

$$C_y = (0)(2\sqrt{3}) - (6)(0) = 0$$

$$C_z = (6)(2) - (0)(2\sqrt{3}) = 12$$

Thus again we have $\vec{C} = 12\hat{k}$.



EVALUATE: Both methods give the same result. Depending on the situation, one or the other of the two approaches may be the more convenient one to use.

Test Your Understanding of Vector Multiplications

Vector $\vec{A}$ has magnitude 2 and vector $\vec{B}$ has magnitude 3. The angle ϕ between $\vec{A}$ and $\vec{B}$ is known to be 0° , 90° , or 180° . For each of the following situations, state what the value of ϕ must be. (In each situation, there may be more than one correct answer.) (a) $\vec{A} \cdot \vec{B} = 0$ (b) $\vec{A} \times \vec{B} = 0$ (c) $\vec{A} \cdot \vec{B} = 6$
(d) $\vec{A} \cdot \vec{B} = -6$ (e) (Magnitude of $\vec{A} \times \vec{B} = 6$

1. Can you find two vectors with different lengths that have a vector sum of zero? What length restrictions are required for three vectors to have a vector sum of zero? Explain your reasoning.
2. One sometimes speaks of the “direction of time,” evolving from past to future. Does this mean that time is a vector quantity? Explain your reasoning.
3. Air traffic controllers give instructions to airline pilots, telling them in which direction they are to fly. These instructions are called “vectors.” If these are the only instructions given, in the name “vector” used correctly? Why or why not?
4. Can you find a vector quantity that has a magnitude of zero but components that are different from zero?
5. Explain. Can the magnitude of a vector be less than the magnitude of any of its components? Explain.
6. Does it make sense to say that a vector is negative? Why? (b) Does it make sense to say that one vector is the negative of another? Why? Does your answer here contradict what you said in part (a)?

Projectile Motion

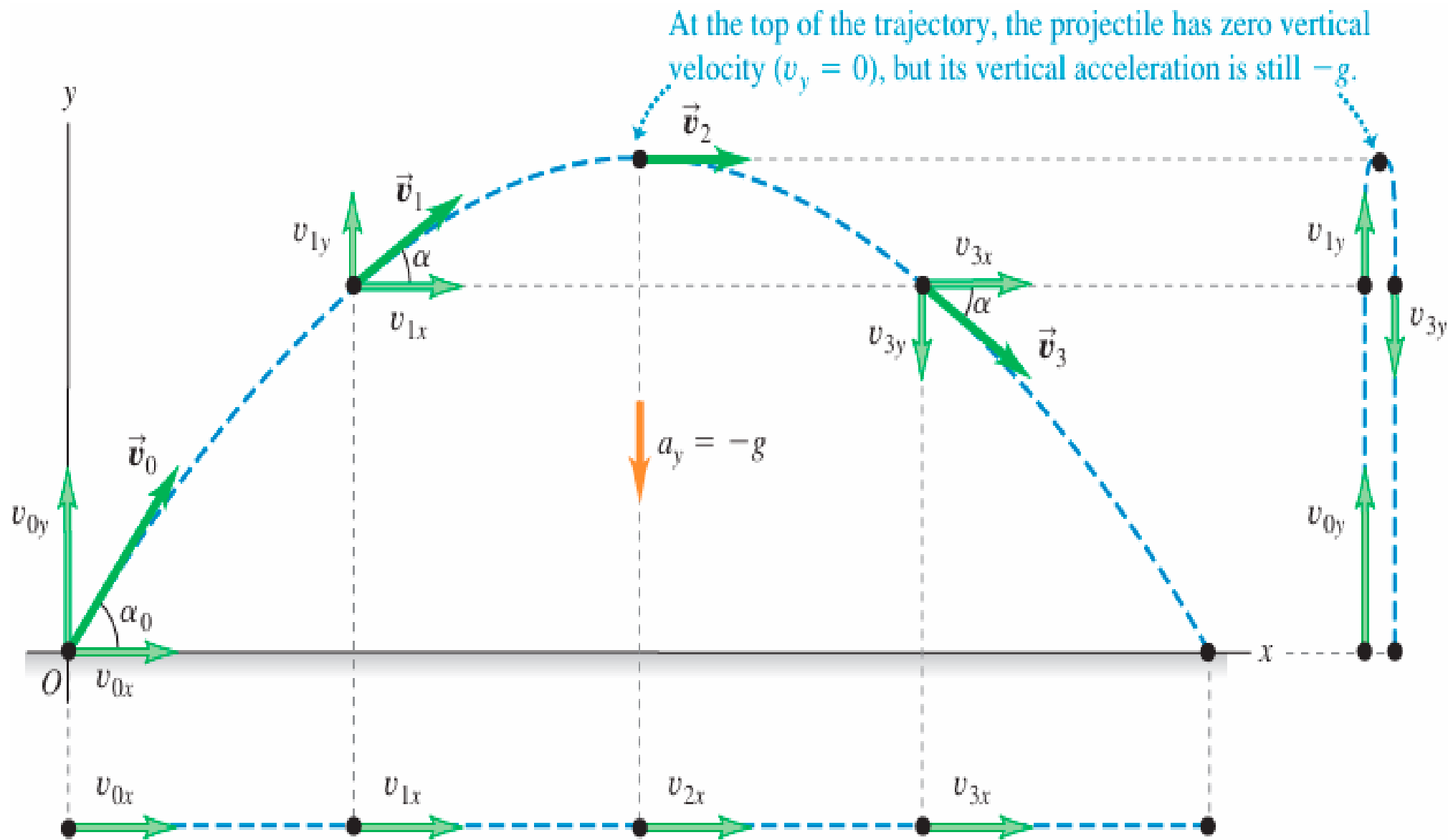
Projectile motion is the motion of an object that is thrown or projected into the air and then moves only under the influence of gravity (air resistance neglected).

Examples:

- A ball thrown at an angle
- A stone thrown horizontally from a building
- A bullet fired from a gun
- A football kicked into the air
- Once the object is projected, no force acts on it except gravity.

Basic Assumptions in Projectile Motion

- Air resistance is negligible
- Acceleration due to gravity, g is constant
- The Earth is flat over the range of motion
- Motion takes place near the surface of the Earth.



At the top of the trajectory, the projectile has zero vertical velocity ($v_y = 0$), but its vertical acceleration is still $-g$.

Vertically, the projectile is in constant-acceleration motion in response to the earth's gravitational pull. Thus its vertical velocity *changes* by equal amounts during equal time intervals.

Horizontally, the projectile is in constant-velocity motion: Its horizontal acceleration is zero, so it moves equal x -distances in equal time intervals.

$$a_x = 0 \quad a_y = -g \quad (\text{projectile motion, no air resistance})$$

$$v_x = v_{0x}$$

$$x = x_0 + v_{0x}t$$

For the y-motion we substitute y for x, v_y for v_x , v_{0y} for v_{0x} , and $a_y = -g$ for a_x :

$$v_y = v_{0y} - gt$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$v_{0x} = v_0 \cos \alpha_0 \quad v_{0y} = v_0 \sin \alpha_0$$

$$x = (v_0 \cos \alpha_0)t \quad (\text{projectile motion})$$

$$y = (v_0 \sin \alpha_0)t - \frac{1}{2}gt^2 \quad (\text{projectile motion})$$

$$v_x = v_0 \cos \alpha_0 \quad (\text{projectile motion})$$

$$v_y = v_0 \sin \alpha_0 - gt \quad (\text{projectile motion})$$

$$r = \sqrt{x^2 + y^2}$$

$$v = \sqrt{v_x^2 + v_y^2}$$

$$\tan \alpha = \frac{v_y}{v_x}$$

$$y = (\tan \alpha_0)x - \frac{g}{2v_0^2 \cos^2 \alpha_0}x^2$$

$$y = bx - cx^2$$

Analysis of Projectile Motion (Key Idea)

- Projectile motion can be divided into two independent motions:
 - Horizontal motion
 - Vertical motion

These two motions occur simultaneously but are independent of each other, connected only by time.

Horizontal Motion of a Projectile

There is no acceleration in the horizontal direction (if air resistance is neglected).

Horizontal velocity remains constant throughout the motion.

Horizontal velocity:

If the projectile is thrown with initial velocity $u_x = u \cos \theta$

Horizontal displacement:

$$X = u \cos \theta t$$

Horizontal motion is uniform motion

Follows the laws of motion with constant velocity.

Vertical Motion of a Projectile

- Vertical motion is affected by gravity.
- Acceleration is always downward and equal to g .

Initial vertical velocity:

$$u_y = u \sin \theta$$

Vertical velocity at time t . $v_y = u \sin \theta - gt$

Vertical displacement: $y = u \sin \theta t - \frac{1}{2}gt^2$

So:

- Vertical motion is uniformly accelerated motion.
- Same as motion under gravity (free fall).

Equation of trajectory

$$x = u \cos \theta \times t \Rightarrow t = \frac{x}{u \cos \theta} \quad \dots(i)$$

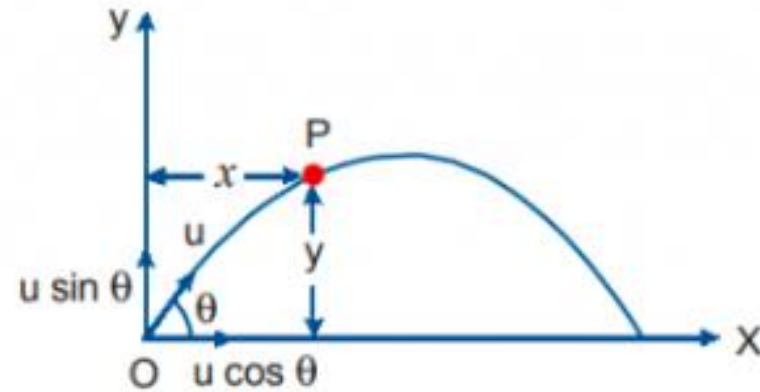
For vertical motion,

$$y = (u \sin \theta)t - \frac{1}{2}gt^2 \quad \dots(ii)$$

$$= u \sin \theta \left(\frac{x}{u \cos \theta} \right) - \frac{1}{2}g \left(\frac{x^2}{u^2 \cos^2 \theta} \right)$$

From equation (i) and (ii)

$$y = x \tan \theta - \frac{1}{2} \frac{gx^2}{u^2 \cos^2 \theta}$$



This equation shows that the trajectory of projectile is parabolic because it is similar to equation of parabola

$$y = ax - bx^2$$

Force

The force is an important factor in the field of Mechanics, which may be broadly *defined as *an agent which produces or tends to produce, destroys or tends to destroy motion. e.g., a horse applies force to pull a cart and to set it in motion.* Force is also required to work on a bicycle pump. In this case, the force is supplied by the muscular power of our arms and shoulders.

✓ EFFECTS OF A FORCE

A force may produce the following effects in a body, on which it acts :

1. It may change the motion of a body. i.e. if a body is at rest, the force may set it in motion. And if the body is already in motion, the force may accelerate it.
2. It may retard the motion of a body.
3. It may retard the forces, already acting on a body, thus bringing it to rest or in equilibrium.
4. It may give rise to the internal stresses in the body, on which it acts.

CHARACTERISTICS OF A FORCE

In order to determine the effects of a force, acting on a body, we must know the following characteristics of a force :

1. Magnitude of the force (i.e., 100 N, 50 N, 20 kN, 5 kN, etc.)
2. The direction of the line, along which the force acts (i.e., along OX, OY, at 30° North of East etc.). It is also known as line of action of the force.
3. Nature of the force (i.e., whether the force is push or pull). This is denoted by placing an arrowhead on the line of action of the force.
4. The point at which (or through which) the force acts on the body.

- ✓ A **particle** may be defined as a body of infinitely small volume and is considered to be a concentrated point.
- ✓ A **rigid body** may be defined as a body that can retain its shape and size, even if subjected to some external forces. In actual practice, no body is perfectly rigid. But for the sake of simplicity, we take all the bodies as rigid bodies.

Basic (Fundamental) Forces in Nature

In nature, **all physical interactions** can be explained using **four fundamental forces**.

These are called **basic forces of nature**.

The four basic forces of nature are

1. Gravitational,
2. Electromagnetic,
3. Strong nuclear, and
4. Weak nuclear forces.

which together explain all physical interactions in the universe.



Graviton?

Solar systems
Galaxies

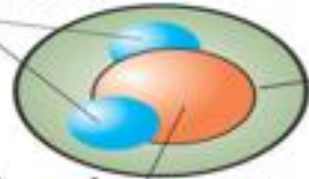


Gravity Force



Electromagnetic force

Hydrogen atom



Water molecule

Oxygen atom

Protons and
Neutrons

Oxygen atom

Electron



Photon



Atoms
Light
Chemistry
Electronics



Strong force

Gluons (8)



Quarks



Mesons
Baryons



Nuclei

Weak force



Bosons (W,Z)



Neutron decay
Beta decay
Neutrino interactions
Burning of the sun

Gravitational Force

The force of attraction between **any two masses** in the universe.

Formula (Newton's Law of Gravitation)

$$F = G \frac{m_1 m_2}{r^2}$$

✓ Characteristics

- Always **attractive**
- Acts between all masses
- **Weakest** of all forces
- **Infinite range**

✓ Effects

- Falling of objects
- Motion of planets
- Tides
- Weight of bodies

✓ Example

- Earth attracting a stone

Electromagnetic Force

The force between **charged particles**.

✓ Characteristics

- Can be **attractive or repulsive**
- Much **stronger than gravity**
- Acts between electric charges
- **Infinite range**

✓ Effects

- Electric forces
- Magnetic forces
- Light (electromagnetic waves)
- Chemical bonding

✓ Example

- Attraction between electron and proton in an atom

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$$

Strong Nuclear Force

The force that holds protons and neutrons together in the nucleus.

✓ Characteristics

- Strongest force in nature
- Acts only at very short range ($\sim 10^{-15}$ m)
- Overcomes repulsion between protons

✓ Effects

- Stability of atomic nuclei
- Nuclear energy

✓ Example

Binding of nucleons inside the nucleus

Weak Nuclear Force

The force responsible for **radioactive decay**.

✓ Characteristics

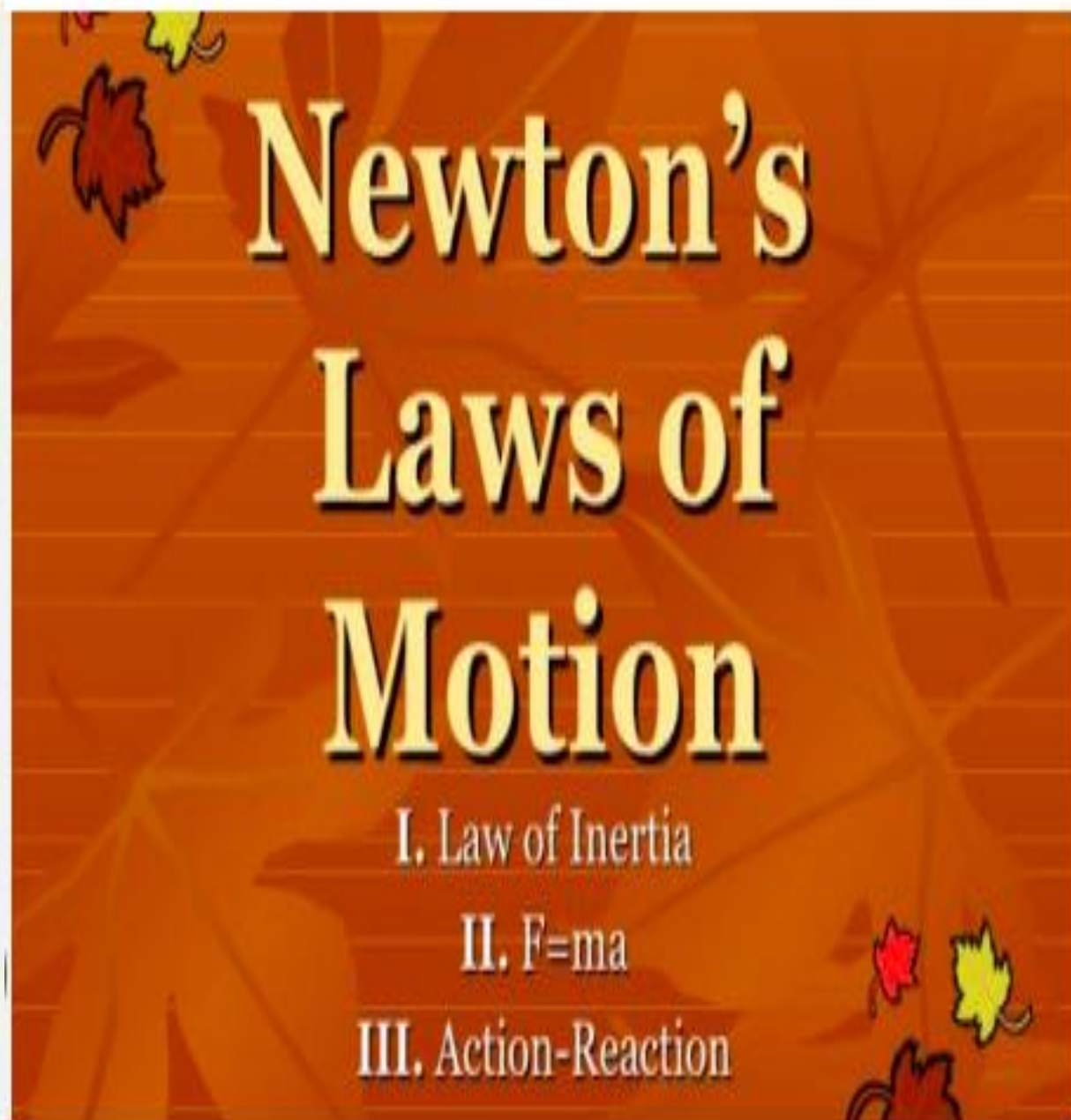
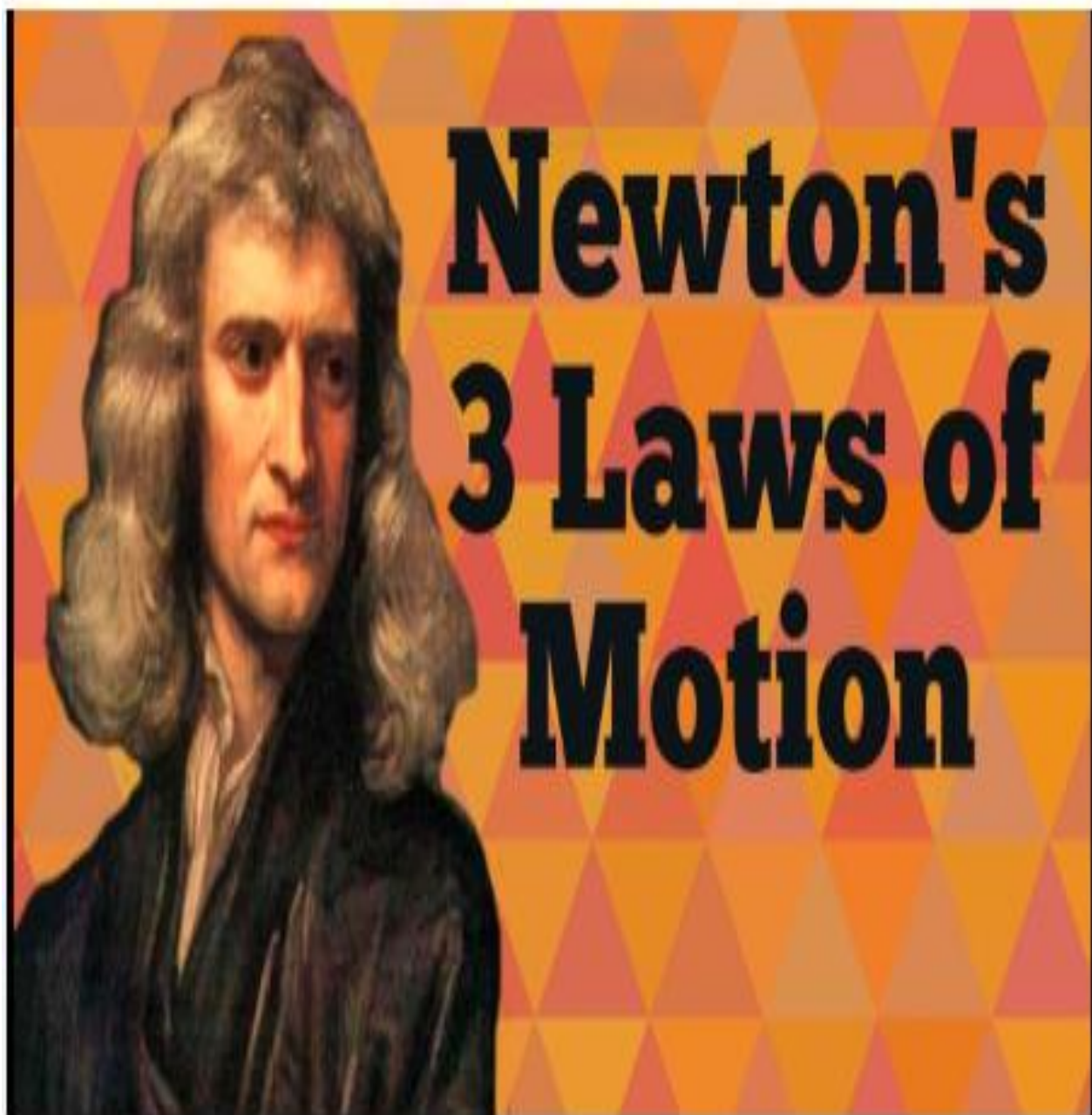
- Very **short range**
- Weaker than the strong force but stronger than gravity
- Acts on subatomic particles

✓ Effects

- Beta decay
- Nuclear reactions in stars

✓ Example

Decay of a neutron into a proton, an electron, and an antineutrino



Newton's first law of motion:

- Motion depends on the net force acting on a body.
- If the net force is zero, a body at rest remains at rest.
- A body in motion with zero net force continues moving at constant speed.
- Objects stop moving in everyday life due to friction, not because motion naturally ends.
- Greater friction causes an object to slow down sooner (tabletop > ice > air-hockey table).
- With no friction, an object would keep moving without any applied force.
- Therefore, a force is not required to sustain motion, only to change it.

Newton's first law of motion: A body acted on by no net force moves with constant velocity (which may be zero) and zero acceleration.

Inertia

- Inertia is the property of a body that resists changes in its motion.
- Due to inertia, a moving object tends to keep moving once set in motion.
- Shaking a ketchup bottle works because the ketchup continues moving forward when the bottle is jerked back.
- Inertia also causes an object at rest to remain at rest unless acted on by a force.
- Pulling a tablecloth quickly does not move the chain much because its inertia resists motion.
- Newton's first law depends on the net force, not individual forces.
- A book resting on a table has two forces acting on it.
- Downward gravitational force from Earth.
- These two forces are equal and opposite, so the net force is zero.
- With zero net force, the book remains at rest, as predicted by Newton's first law.
- The same idea applies to a hockey puck on a frictionless surface.
- A body is said to be in equilibrium when it is at rest or moving with constant velocity.
- For equilibrium, the net force must be zero.
- This requires that the net force in each direction is zero.

Conceptual Example 4.2 Zero net force means constant velocity

In the classic 1950 science fiction film *Rocketship X-M*, a spaceship is moving in the vacuum of outer space, far from any star or planet, when its engine dies. As a result, the spaceship slows down and stops. What does Newton's first law say about this scene?

SOLUTION

After the engine dies there are no forces acting on the spaceship, so according to Newton's first law it will *not* stop but will continue to move in a straight line with constant speed. Some science fiction movies are based on accurate science; this is not one of them.

Conceptual Example 4.3 Constant velocity means zero net force

You are driving a Maserati GranTurismo S on a straight testing track at a constant speed of 250 km/h. You pass a 1971 Volkswagen Beetle doing a constant 75 km/h. On which car is the net force greater?

SOLUTION

The key word in this question is “net.” Both cars are in equilibrium because their velocities are constant; Newton's first law therefore says that the *net* force on each car is *zero*.

This seems to contradict the “common sense” idea that the faster car must have a greater force pushing it. Thanks to your

Maserati's high-power engine, it's true that the track exerts a greater forward force on your Maserati than it does on the Volkswagen. But a *backward* force also acts on each car due to road friction and air resistance. When the car is traveling with constant velocity, the vector sum of the forward and backward forces is zero. There is more air resistance on the fast-moving Maserati than on the slow-moving Volkswagen, which is why the Maserati's engine must be more powerful than that of the Volkswagen.

Newton's Second law

- A constant net force produces constant acceleration.
- When the net force acts in the direction of motion, the object's speed increases.
- In this case, acceleration, velocity, and net force all point in the same direction.
- When the net force acts opposite to the direction of motion, the object slows down.
- The acceleration always points in the direction of the net force, not necessarily the velocity.
- Experiments show that if the net force is constant, the acceleration is also constant.
- A net force causes acceleration in its own direction.
- These ideas also apply to curved motion.
- In uniform circular motion, the net force and acceleration are directed toward the center of the circle.
- Even though the force and acceleration exist, the speed remains constant while the direction of motion changes.

- A **constant horizontal force** is applied to a puck on a **frictionless surface**.
- This applied force is equal to the **net force** on the puck.
- Changing the **magnitude of the net force** changes the **acceleration**.
- **Doubling the net force doubles the acceleration.**
- **Halving the net force halves the acceleration.**
- Experiments show that for a given body, **acceleration is directly proportional to the net force** acting on it.

$$\sum F_x = ma_x \quad \sum F_y = ma_y \quad \sum F_z = ma_z \quad \text{(Newton's second law of motion)}$$

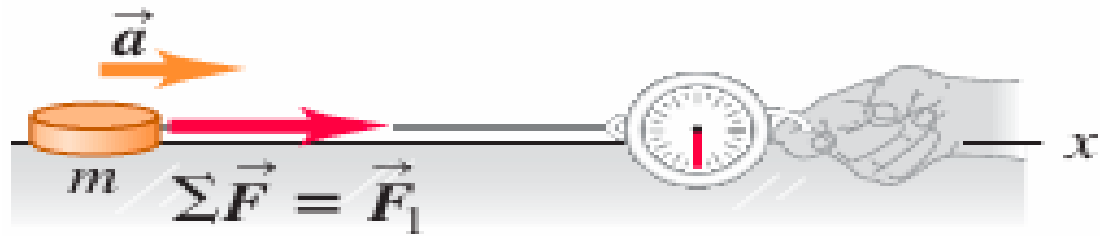
Mass and Force

Our results mean that for a given body, the *ratio* of the magnitude $|\sum \vec{F}|$ of the net force to the magnitude $a = |\vec{a}|$ of the acceleration is constant, regardless of the magnitude of the net force. We call this ratio the *inertial mass*, or simply the **mass**, of the body and denote it by m . That is,

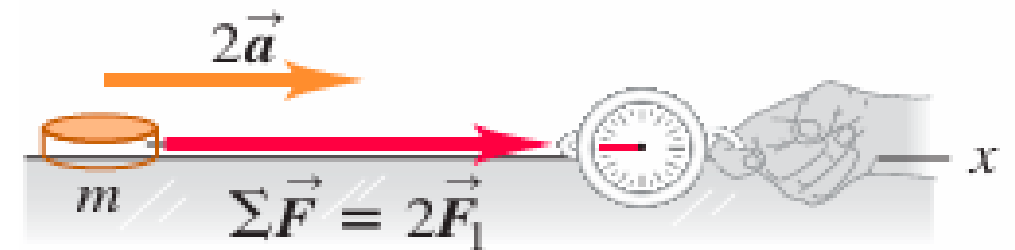
$$m = \frac{|\sum \vec{F}|}{a} \quad \text{or} \quad |\sum \vec{F}| = ma \quad \text{or} \quad a = \frac{|\sum \vec{F}|}{m} \quad (4.5)$$

For a body of a given mass m , the magnitude of the body's acceleration is directly proportional to the magnitude of the net force acting on the body.

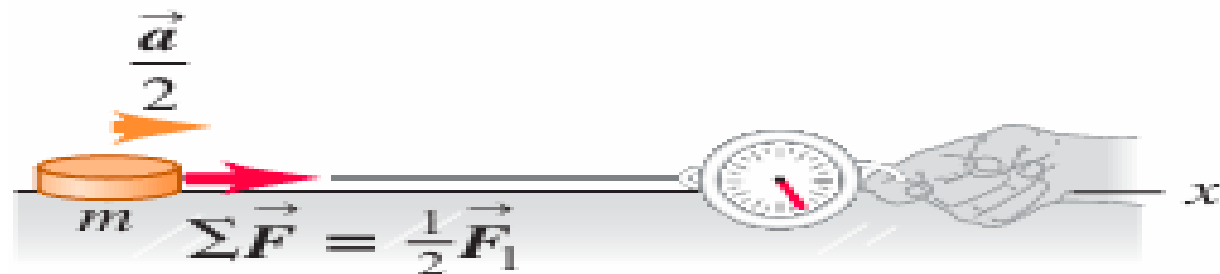
(a) A constant net force $\Sigma \vec{F}$ causes a constant acceleration $\vec{a}$.



(b) Doubling the net force doubles the acceleration.



(c) Halving the force halves the acceleration.



Newton's second law of motion: If a net external force acts on a body, the body accelerates. The direction of acceleration is the same as the direction of the net force. The mass of the body times the acceleration of the body equals the net force vector.

$$\sum \vec{F} = m\vec{a} \quad (\text{Newton's second law of motion})$$

$$\vec{a} = \frac{\sum \vec{F}}{m}$$

4.17 The design of high-performance motorcycles depends fundamentally on Newton's second law. To maximize the forward acceleration, the designer makes the motorcycle as light as possible (that is, minimizes the mass) and uses the most powerful engine possible (thus maximizing the forward force).



CAUTION $m\vec{a}$ is not a force You must keep in mind that even though the vector $m\vec{a}$ is equal to the vector sum $\sum \vec{F}$ of all the forces acting on the body, the vector $m\vec{a}$ is *not* a force. Acceleration is a *result* of a nonzero net force; it is not a force itself. It’s “common sense” to think that there is a “force of acceleration” that pushes you back into your seat when your car accelerates forward from rest. But *there is no such force*; instead, your inertia causes you to tend to stay at rest relative to the earth, and the car accelerates around you (see Fig. 4.11a). The “common sense” confusion arises from trying to apply Newton’s second law where it isn’t valid, in the noninertial reference frame of an accelerating car. We will always examine motion relative to *inertial* frames of reference only. ■

Some Notes on Units

A few words about units are in order. In the cgs metric system (not used in this book), the unit of mass is the gram, equal to 10^{-3} kg, and the unit of distance is the centimeter, equal to 10^{-2} m. The cgs unit of force is called the *dyne*:

$$1 \text{ dyne} = 1 \text{ g} \cdot \text{cm/s}^2 = 10^{-5} \text{ N}$$

In the British system, the unit of force is the *pound* (or pound-force) and the unit of mass is the *slug* (Fig. 4.20). The unit of acceleration is 1 foot per second squared, so

$$1 \text{ pound} = 1 \text{ slug} \cdot \text{ft/s}^2$$

The official definition of the pound is

$$1 \text{ pound} = 4.448221615260 \text{ newtons}$$

Mass and Weight

- Although **mass** and **weight** are related, they are **not the same**. Understanding their difference is very important in physics.

Sl. No.	Mass	Weight
1.	The mass is a scalar quantity.	The weight is a vector quantity.
2.	Mass of a rigid body is regular everywhere in the universe.	The weight of a rigid body alters from place to place and inclines zero at the center of the earth.
3.	Mass can be resulted by a traditional balance.	Weight can be defined as spring balance
4.	The unit of mass is kg or g.	The unit of weight is Newton.
5.	Mass can never be zero.	Weight can be zero based on the gravity acting upon it.
6.	Mass does not change based on location.	Weight changes based on location, depending on the gravity it experiences.
7.	Mass is measured using an ordinary weighing scale.	Weight is measured using spring balance.

Friction

Friction is a resistive force opposing motion between surfaces, acting proportionally to the normal force and influenced by surface texture. Drag force acts similarly within fluids, increasing with speed ($F_D \propto v^2$)

Terminal speed is the maximum constant velocity achieved when drag force equals the gravitational force.

Friction and Its Properties Definition:

A resistive force that opposes the relative motion of two surfaces in contact.

Surface Dependence: Depends on the nature of the surfaces (roughness) and the normal force (F_N), not typically on surface area.

Formula: $F_f = \mu F_N$, where μ is the coefficient of friction.

Types: Static friction (prevents motion) and kinetic friction (opposes moving objects).

(1) Static Friction

(2) Kinetic Friction

(3) Rolling Friction

(4) Sliding Friction

Practice Problems on Frictional Force

1. A 1000 kg car is parked on a flat road. If the coefficient of static friction between the tires and the road is 0.75, what is the maximum force that can be applied horizontally to the car before it starts to move?
2. A 50 kg box is sliding on a horizontal surface with a coefficient of kinetic friction of 0.2. If a horizontal force of 100 N is applied to the box, what will be its acceleration?
3. A 200 g book is initially resting on a wooden table with a coefficient of static friction of 0.4. If the table is gradually tilted, at what angle (in degrees) will the book start to slide?
4. A 500 N block is placed on an inclined plane with an angle of 30 degrees. The coefficient of kinetic friction between the block and the plane is 0.3. Calculate the net force acting on the block parallel to the inclined plane if it is moving at a constant velocity.
5. A 10 kg block is hanging from a rope over a pulley. The coefficient of kinetic friction between the block and the surface is 0.2. If the block is released from rest, what will its acceleration be?